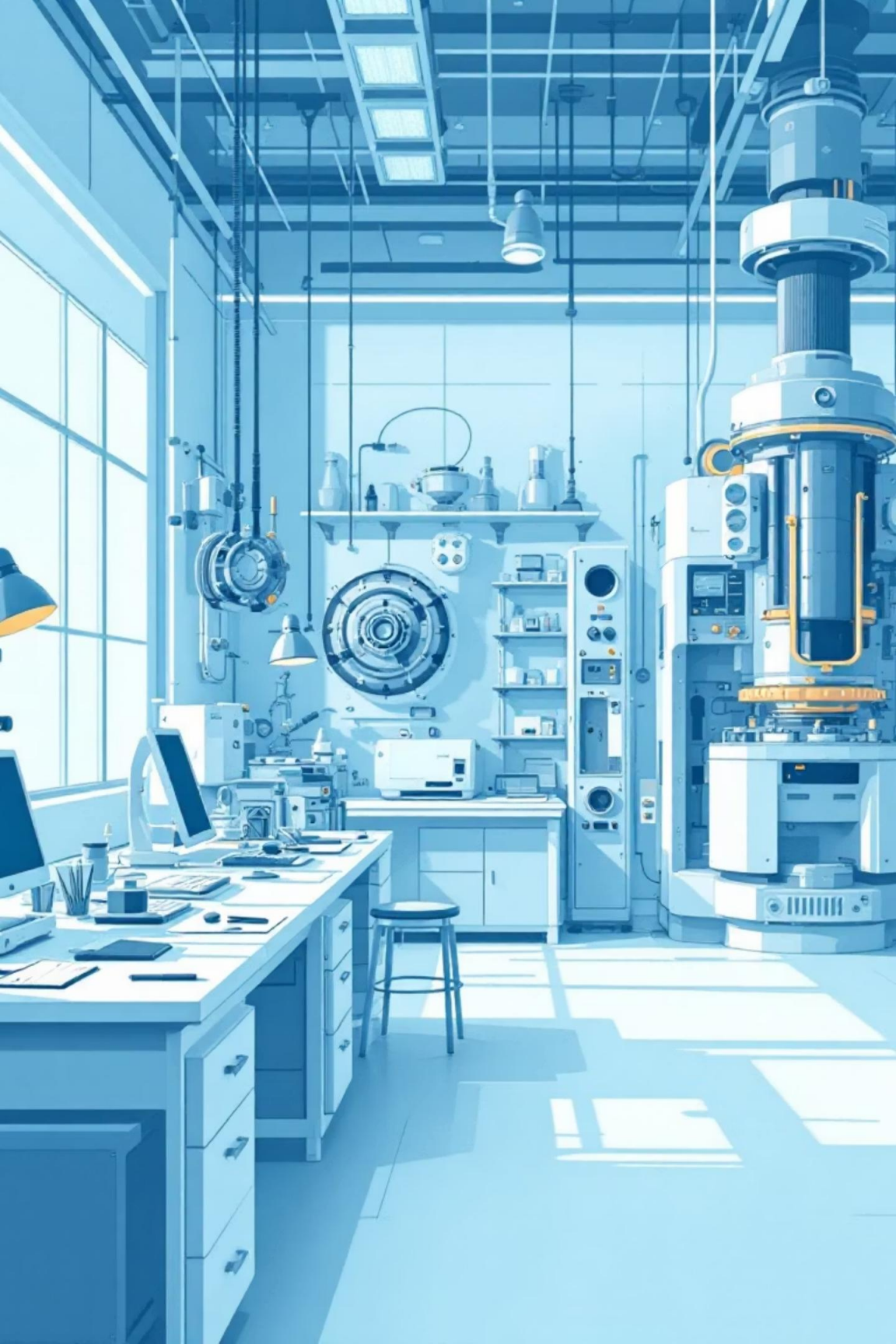




# Mechanical Integrator

## TA212 Lab Project Report

Group 9 | Tuesday Lab

# Project Overview

A comprehensive mechanical integrator project combining precision machining, 3D printing, and gear systems. Led by **Dr. Sarvesh Mishra** with guidance from **Arun Kumar Dubey**.

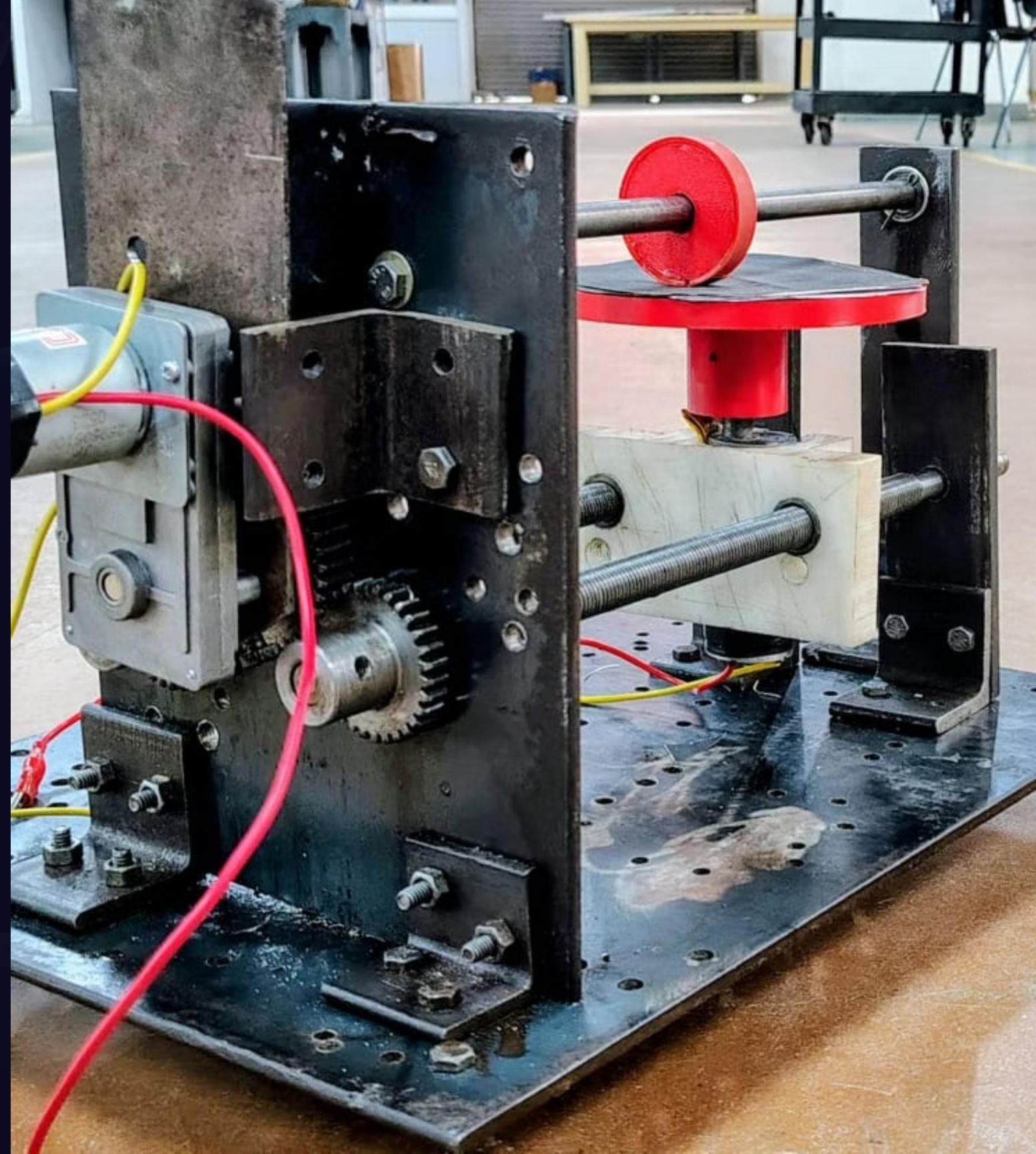
## Team Members

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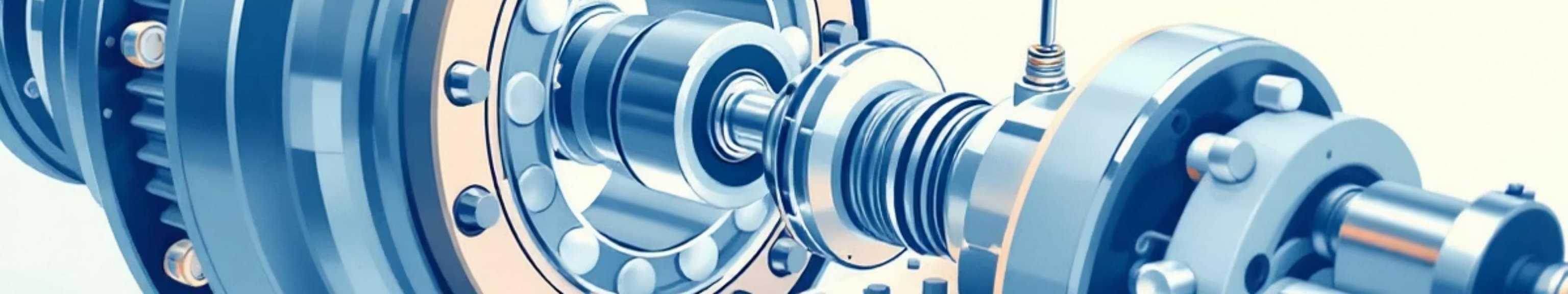


# Complete Assembly

The mechanical integrator comprises 15 precision components working in harmony. Key elements include ball bearings, drive shafts, spur gears, and a carriage system designed for smooth rotational motion and linear translation.







# Core Components

1

## Stationary Rod

1 unit, mild steel. Provides fixed reference axis for the system.

2

## Drive Shaft

2 units, mild steel. Transmits rotational power through the mechanism.

3

## Gear System Design

The integrator uses precision spur gears to transmit motion. Two 30-tooth gears mesh with one 40-tooth gear creating a gear ratio system with module 1.5.

# Structural Elements

## Support Stands

2 units, mild steel. Provides rigid framework and stability.

## Carriage/Mover

1 unit, Derlin. Enables linear motion along the Drive Shaft.

## Base Disk

1 unit, PLA. 3D-printed component, which rotates with a constant  $\omega$  and gives a range of values for our Function.

## Small Disk

1 unit, PLA. 3D-printed component which tracks the number of rotation over time.





# Manufacturing Processes

01

## Conventional Machining

Turning, lathing, drilling, and milling for steel components. Tooth cutting for gear fabrication using indexing methods.

02

## 3D Printing

PLA material used for base disk and stationary wheel. Enables complex geometries and rapid prototyping.

03

## Assembly Integration

Precision fitting of components with nut-bolt and taper pins. Quality control and dimensional verification.

04

## Automation and Control

The mechanical system was automated using an Arduino-based control unit. This managed all electronic components, including the motors responsible for driving the system inputs.

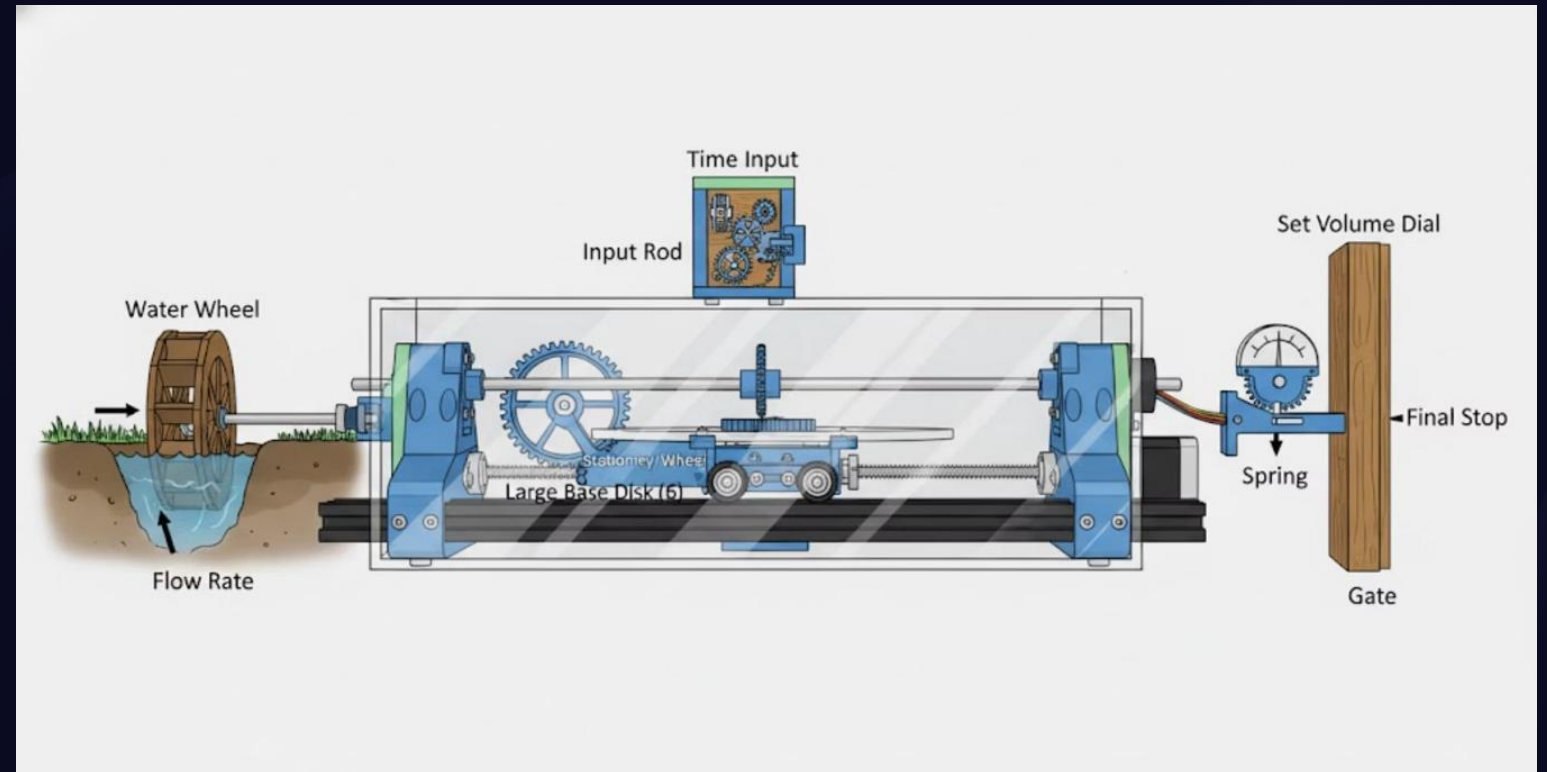


# Uses of Mechanical Integrator in Real Life

Our project aimed to solve critical problems associated with traditional furrow irrigation, a widely used method in agriculture that often leads to significant water waste and uneven crop yield.



Furrow Irrigation



## The Challenge: Water Waste & Uneven Distribution

Manual furrow irrigation is inefficient, resulting in up to 30-50% water loss through runoff and deep percolation. This leads to inconsistent moisture levels across fields, impacting crop health and overall yield.

## Our Objective: Precision Control

We designed the mechanical integrator to achieve automated, precise water flow control, targeting a 20% reduction in water usage and a 15% improvement in distribution uniformity. The goal was to ensure optimal water delivery at the plant root zone.



# Other Applications

Mechanical integrators, while often found in complex industrial settings, also have a rich history and contemporary relevance in educational and historical contexts.

## Educational Tool for Calculus

Historically, mechanical integrators were used as tangible teaching aids to demonstrate complex calculus concepts to students. They provided a physical model for understanding integration, making abstract mathematical principles more accessible and intuitive for school-aged children and engineering students alike.

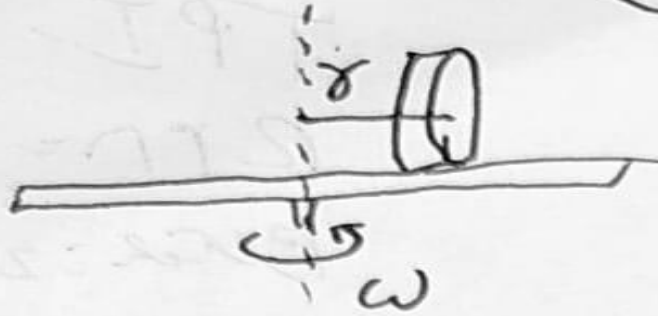
## Ship & Aircraft Navigation

Mechanical integrators were essential for "dead reckoning" navigation, allowing ships and aircraft to track their position without landmarks. Since a vessel's speed is always changing, the integrator solved the difficult problem of calculating total distance. It continuously took a "Time" input from a clock and a "Velocity" input from a speed sensor. By mechanically integrating the changing speed over time, its output shaft turned by an amount equal to the *total distance* traveled, allowing navigators to accurately pinpoint their position on a map.



# Challenges Faced

1. A significant challenge emerged during the assembly of the Support Stand and the Carriage. In our initial design, we discovered a critical mechanical interference where the Drive Shaft would bind, or get stuck, against both the carriage and the stands. This completely restricted the shaft's rotation and forced a comprehensive redesign of the interfering components to achieve the necessary operational clearance.
2. While creating Drive shaft, initially we tried to make the threadings manually but we faced many issue due to long length of the shaft, so then we used lathe machine to make the threads.
3. A significant challenge was the compact and highly integrated nature of the design. The dimensions of each component were interdependent, necessitating a meticulous, iterative process of trial and error to achieve the precise fits and functionality required.



$$\omega r = \omega' R$$

$$\omega' = \frac{\omega r}{R}$$

$r(t) \Rightarrow$  Feed Function

30 rpm

$$\omega = \pi \text{ rad/s}$$

$$\frac{30}{60} \times 2\pi \uparrow$$

$$\int d\theta = \frac{\omega}{R} \int r(t) dt$$

$$2\pi \cdot n = \frac{\omega}{R} \int r(t) dt$$

$$2\pi \cdot n \cdot R = \int r(t) dt$$

$$2\pi \cdot n \cdot \left( \frac{55 \times 10^3}{2} \right) = \int r(t) dt$$

$$n \cdot (55 \times 10^3) = I$$







**THANK YOU!**